FUNCTORIALITY OF THE CECH-TO-COHOMOLOGY DIAGRAM ALONG A MORPHISM OF RINGED SPACES

THE STACKS PROJECT AUTHORS

Editorial note. Preamble: only the retained preamble of this chapter (sources/stacks-cohomology/retained-preamble.tex) is taken, because the chapter file itself carries just an input of the project's single shared preamble on its first line. It holds the Xy-pic setup, the author's native theorem declarations and the mathematical macros, copied line for line; nothing was added, removed or redefined. This card uses neither stix nor transparent, which the host TeX Live lacks. The Xy-pic matrix diagrams in the statement and in the proof rely on the xy package loaded there, which matches the two-pass local trial compilation.

Editorial note. All five external references are pointers inside this chapter (one lemma, three equations, one remark) and serve to resolve the original reference commands; the arrow gamma, the four double complexes and the verification of the whole diagram are extracted verbatim inside this span and are not replaced by those pointers.

Editorial note. Original numbering: in this chapter lemmas, equations and remarks share the subsection counter, and material-only assembly locates passages by label, so the printed numbers need not agree with the original ones.

Editorial note. Uniform work/source_title across every card of this source, as section 11 of NEW_TASK_AUTHORING.md requires (the manifest field work is taken from source_title). The precise section of each claim is carried by claim_key and by statement_anchor in the registration, and the section name is repeated in the editorial notes where it matters.

Allowed standard prerequisites (editorial).

- The comparison map from the Cech complex of a covering to derived global sections. Same chapter, Lemma labelled lemma-cech-cohomology (canonical Stacks tag 01EQ, source line 1434), named in the counted statement at lines 2540 and 2542.. Supplies the horizontal arrows of the two displayed diagrams; the counted content is that the vertical arrow gamma built in the proof makes them commute.
- Pushforward and pullback of modules along a morphism of ringed spaces, and the notion of an f-map f^*G -> F. Standard; fixed in the same chapter's section Cech cohomology and cohomology (source lines 2340 onwards) together with the displays equation-functorial-derived (tag 01F9) and equation-functorial (tag 01FA).. Fixes the objects the diagram is built from and the arrows its sides are made of.
- Injective resolutions and double complexes computing derived global sections. Used at proof lines 2565-2591 as in the proof of Lemma lemma-cech-cohomology (tag 01EQ) and Remark remark-explain-arrow (tag 01FB, source line 2391).. Provides the four double complexes whose morphisms are compared.

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Source reference. Original lemma 01EQ: sources/stacks-cohomology/pinned-original-source.tex, source lines 1434–1434. Same-chapter pointer named in the counted statement (lines 2540 and 2542) and used in the proof (lines 2568 and 2619) for the horizontal arrows of the displayed diagrams, i.e. for the map from the Cech complex to derived global sections. It fixes the arrows that the counted diagram compares; no part of the comparison is taken from it.

Source reference. Original equation 01F9: sources/stacks-cohomology/pinned-original-source.tex, source lines 2370–2370. Same-chapter pointer to the display naming the right vertical arrow of the first diagram in the counted statement (line 2541). Navigation to a display only.

Source reference. Original equation 01FA: sources/stacks-cohomology/pinned-original-source.tex, source lines 2386–2386. Same-chapter pointer to the display giving the right vertical arrow of the cohomology diagram in the second half of the counted statement (line 2552). Navigation to a display only.

Source reference. Original remark 01FB: sources/stacks-cohomology/pinned-original-source.tex, source lines 2391–2391. Same-chapter pointer used at proof lines 2578-2582, where the auxiliary injective resolution f_*I -> (J')^bullet and the morphism beta are chosen exactly as explained in that remark. It points at a construction recipe; the counted proof still has to keep the four double complexes coherent.

Source reference. Original equation 01FC: sources/stacks-cohomology/pinned-original-source.tex, source lines 2405–2405. Same-chapter pointer used at proof lines 2582 and 2621: commutativity of the pentagon in the displayed diagram is derived from commutativity of this chosen square. Navigation to the displayed commutativity that the construction must satisfy.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of ringed spaces. Let $\varphi : f^*\mathcal{G} \to \mathcal{F}$ be an f-map from an $\mathcal{O}_Y$-module $\mathcal{G}$ to an $\mathcal{O}_X$-module $\mathcal{F}$. Let $\mathcal{U} : X = \bigcup_{i\in I} U_i$ and $\mathcal{V} : Y = \bigcup_{j\in J} V_j$ be open coverings. Assume that $\mathcal{U}$ is a refinement of $f^{-1}\mathcal{V} : X = \bigcup_{j\in J} f^{-1}(V_j)$. In this case there exists a commutative diagram*

$$\begin{array}{ccc} \check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{F}) & \longrightarrow & R\Gamma(X, \mathcal{F}) \\ {\scriptstyle\gamma}\uparrow & & \uparrow \\ \check{\mathcal{C}}^\bullet(\mathcal{V}, \mathcal{G}) & \longrightarrow & R\Gamma(Y, \mathcal{G}) \end{array}$$

in $D^+(\mathcal{O}_X(X))$ with horizontal arrows given by Lemma 01EQ and right vertical arrow by (01F9). In particular we get commutative diagrams of cohomology groups

$$\begin{array}{ccc} \check{H}^p(\mathcal{U}, \mathcal{F}) & \longrightarrow & H^p(X, \mathcal{F}) \\ {\scriptstyle\gamma}\uparrow & & \uparrow \\ \check{H}^p(\mathcal{V}, \mathcal{G}) & \longrightarrow & H^p(Y, \mathcal{G}) \end{array}$$

where the right vertical arrow is (01FA)

Proof. We first define the left vertical arrow. Namely, choose a map $c : I \to J$ such that $U_i \subset f^{-1}(V_{c(i)})$ for all $i \in I$. In degree p we define the map by the rule

$$\gamma(s)_{i_0\ldots i_p} = \varphi(s)_{c(i_0)\ldots c(i_p)}$$

This makes sense because φ does indeed induce maps $\mathcal{G}(V_{c(i_0)\ldots c(i_p)}) \to \mathcal{F}(U_{i_0\ldots i_p})$ by assumption. It is also clear that this defines a morphism of complexes. Choose

injective resolutions $\mathcal{F} \to \mathcal{I}^\bullet$ on X and $\mathcal{G} \to J^\bullet$ on Y. According to the proof of Lemma 01EQ we introduce the double complexes $A^{\bullet,\bullet}$ and $B^{\bullet,\bullet}$ with terms

$$B^{p,q} = \check{\mathcal{C}}^p(\mathcal{V}, \mathcal{J}^q) \quad \text{and} \quad A^{p,q} = \check{\mathcal{C}}^p(\mathcal{U}, \mathcal{I}^q).$$

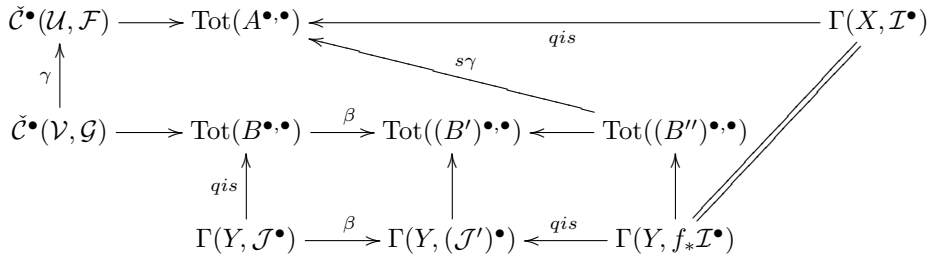
As in Remark 01FB above we also choose an injective resolution $f_*\mathcal{I} \to (\mathcal{J}')^\bullet$ on Y and a morphism of complexes $\beta : \mathcal{J} \to (\mathcal{J}')^\bullet$ making (01FC) commutes. We introduce some more double complexes, namely $(B')^{\bullet,\bullet}$ and $(B'')^{\bullet,\bullet}$ with

$$(B')^{p,q} = \check{\mathcal{C}}^p(\mathcal{V}, (\mathcal{J}')^q) \quad \text{and} \quad (B'')^{p,q} = \check{\mathcal{C}}^p(\mathcal{V}, f_*\mathcal{I}^q).$$

Note that there is an f-map of complexes from $f_*\mathcal{I}^\bullet$ to $\mathcal{I}^\bullet$. Hence it is clear that the same rule as above defines a morphism of double complexes

$$\gamma : (B'')^{\bullet,\bullet} \longrightarrow A^{\bullet,\bullet}.$$

Consider the diagram of complexes



The two horizontal arrows with targets $\mathrm{Tot}(A^{\bullet,\bullet})$ and $\mathrm{Tot}(B^{\bullet,\bullet})$ are the ones explained in Lemma 01EQ. The left upper shape (a pentagon) is commutative simply because (01FC) is commutative. The two lower squares are trivially commutative. It is also immediate from the definitions that the right upper shape (a square) is commutative. The result of the lemma now follows from the definitions and the fact that going around the diagram on the outer sides from $\check{\mathcal{C}}^\bullet(\mathcal{V}, \mathcal{G})$ to $\Gamma(X, \mathcal{I}^\bullet)$ either on top or on bottom is the same (where you have to invert any quasi-isomorphisms along the way). □